Gujarat Technological University

Diploma in Engineering — Modern Physics

Unit 2: Electrostatics

Topic 1: Electric Charge and Coulomb's Law

Definition: Electric charge is a basic property of matter that causes it to feel a force when placed near other matter, and **Coulomb's Law** calculates the push or pull force between two static charges.

3 Features:

- Charges exist in two types: positive (protons) and negative (electrons).
- Like charges repel each other, while opposite charges attract each other.
- Charge is conserved, meaning it cannot be created or destroyed, only transferred.

3 Uses:

- Helps engineers understand how static electricity builds up on moving factory machinery.
- Used to design industrial air filters that trap dust particles using electrical attraction.
- Forms the absolute foundation for studying electrical currents in circuit boards.

Real-Life Application: Rubbing a plastic comb against your hair transfers electrons. The comb gets a static charge and can easily pick up tiny, uncharged pieces of paper.

Topic 2: Electric Field and Field Lines

Definition: An **electric field** is the invisible space around a charged object where its electrical force can be felt by other charges, visualized using paths called **electric field lines**.

3 Features:

- Field intensity (E = F/q) measures how strong the electrical force is at any specific point.
- Field lines always point outward from positive charges and inward toward negative charges.
- Field lines never cross each other because the electrical force can only point in one direction at a time.

3 Uses:

- Used to design old display monitors and cathode-ray tubes by guiding electron paths.

- Helps engineers place electrical grid lines safely by predicting how high voltages affect surrounding air.
- Assists in choosing insulation materials for high-voltage transformers.

Real-Life Application: Photocopy machines use strong electric fields to precisely guide black ink powder onto the paper in the exact shape of your document letters.

Topic 3: Electric Potential and Potential Difference

Definition: Electric potential is the electrical work done to bring a unit charge from far away to a specific point, and **potential difference** (voltage) is the electrical pressure needed to move a charge between two points.

3 Features:

- Measured in Volts ($V = W/q$) and acts like a pump for electricity.
- Electric charges naturally flow from a point of high potential to a point of low potential.
- It relates directly to the electric field strength over a specific distance ($V = E \times d$).

3 Uses:

- Determines how strongly a battery can push electrical current through a circuit.
- Helps technicians select the right voltage rating for household appliances.
- Used to safely manage electric grids and prevent dangerous power surges.

Real-Life Application: A standard 9V battery creates a potential difference of 9 Volts between its terminals, forcing current to flow and power up a toy car's motor.

Topic 4: Capacitance and Parallel Plate Capacitors

Definition: A **capacitor** is an electronic component that stores electrical energy, and **capacitance** ($C = Q/V$) is its ability to hold an electric charge.

3 Features:

- A parallel plate capacitor consists of two metal plates separated by an insulating gap ($C = \varepsilon_0 A / d$).
- Storing capacity increases if the plates are larger or brought closer together.
- Placing an insulating material (dielectric) between the plates increases its overall storage capacity.

3 Uses:

- Filters out unwanted electric noise or ripples in electronic power supplies.
- Stores quick energy for camera flashes that require a sudden, bright burst of light.
- Used in radio tuning circuits to select specific station frequencies.

Real-Life Application: The touchscreens on modern smartphones use tiny capacitors. When your finger touches the glass, it changes the capacitance, telling the phone exactly where you tapped.

Topic 5: Series and Parallel Combination of Capacitors

Definition: Capacitors can be connected together either end-to-end in a single line (**series**) or side-by-side across the same voltage source (**parallel**) to change total storage capacity.

3 Features:

- In a series combination, total capacitance decreases, but the network can handle higher voltages.
- In a parallel combination, total capacitance increases by adding them all together ($C_{total} = C_1 + C_2$).
- Series circuits share voltage, while parallel circuits keep the voltage identical across all components.

3 Uses:

- Allows engineers to create custom capacitance values using standard factory parts.
- Used to bundle energy cells safely inside portable power banks.
- Protects delicate electronics from high-voltage spikes by splitting the electric load.

Real-Life Application: Computer power supplies combine multiple small capacitors in parallel to create a massive storage bank that keeps the computer running during brief power drops.